

COSC 360 Project – User Guide & Walkthrough

1. Introduction

This web application is a full-stack course hub and discussion platform called Together as One. Students and instructors can browse, create, and join course hubs, participate in threaded discussions, share file attachments, and manage their profiles. The system supports JWT-based authentication, role-based access (admin vs. user), real-time status polling, and a fully Dockerized deployment.

2. How to Run the Application

Using Docker (Recommended)

1. Clone the repository
2. Create .env files for both frontend and backend, copy from .example.env files
3. Run docker compose up --build
4. Open the application:
 - a. Frontend: <http://localhost:4000>
 - b. Backend: <http://localhost:3000>

The database is seeded automatically on first startup, no manual setup required.

3. Test Accounts (Alternatively, you can register a new account)

Role	Email	Password	Username
Admin	admin@test.com	@Admin12345678	admin
Regular User	user@test.com	@User12345678	serena
Regular User2	user@test.com	@User12345678	user2
Facebook Login	c07917277@gmail.com	Xuruoyan0!	not needed

4. Walkthrough Steps

Step 1: Access the Landing Page

- Open <http://localhost:4000>
- The Landing Page displays a screen with a “Get Started” button
- Click Get Started to go to the Home page.

Step 2: Home Page

- The Home page shows all available course hubs as cards.
- Each card shows the course title, field, type (e.g., Lecture, Lab), tags, and member count.
- A Spotlight Section highlights featured courses at the top.
- You can favourite courses using the heart icon on each card (stored locally in your browser).

Step 3: Search and Filter Courses

- Use the Search Bar at the top to search for courses by title or keyword.
- Search is asynchronous, results update as you type (300ms debounce, no reload).
- Use the Filter panel to narrow results by:
 - Course prefix (e.g., COSC, MATH, PHYS)

- Level (100–400 Level, Career Development)
- Sort order (A→Z, Z→A, Most Members, Least Members)

Step 4: Register a New Account

- Click Login in the top navigation bar.
- Click the Sign Up link.
- Fill in: Username, Email, Password, Confirm Password.
- Optionally upload a profile image during signup.
- After registration you will be redirected to the login screen.

Step 5: Login

- Click the login button in the navigation bar.
- Enter one of the test accounts, or use the Social Login options (Continue with Google/Facebook) to authenticate via OAuth 2.0.
- After login, the navigation bar will display the username and the user's social profile avatar.

Step 6: Course Overview Page

- From the Home page, click any course card or View Details.
- The Course Overview page shows:
 - Course title, description, tags, field, type, location
 - Member count and application deadline (if set)
 - 3 most recent discussion posts as a preview
- Click Join Hub to become a member of that course (requires login).
- If you are already a member, the button changes to Leave Hub.

Step 7: Discussion System

- From Course Overview, click the Discussion tab (via the ProjectTabs bar).
- All posts for that course are listed in reverse-chronological order.
- Each post shows: author avatar/initial, username, post text, timestamp, and any attached files.
- Create a new post:
 - Type your message in the text box, and optionally click the paperclip to attach up to 5 files.
 - Click Post, the new post appears instantly at the top.
- Reply to a post: Click Reply under any post, type your reply, and submit.
- Delete a post or reply: Click the red icon (visible on your own posts, or all posts if you are admin).

Step 8: Team/Members Page

- Click the Team tab from within a course.
- Shows all members who have joined the course.
- If you are the course creator or admin, a Manage Members button appears to remove members.

Step 9: Dashboard

- Click the Dashboard icon in the Navbar (grid icon).
- Shows all courses you have joined.

- Click a course to navigate back to its overview.
- Click Leave to leave a course.
- A Join Course modal lets you join a new course by searching.

Step 10: Create a New Course Hub (Admin)

- Click the + Create Course button from the Dashboard (available to all logged-in users).
- Fill in: Title, Description, Field, Type/Level, Tags, Location, Start/End Date, Positions.
- Click Create, the new hub appears immediately on the Home page.

Step 11: Edit/Delete a Course Hub (Admin)

- On any Course Overview page, admin users see an ACTIONS panel on the right.
- Click Edit Hub to open an inline modal to update course details.
- Click Delete Hub (red button) to permanently remove the course.

Step 12: User Search & Account Management (Admin)

- In the ACTIONS panel, click Search Users.
- Search by username, email, or even a keyword in post text.
- Each result shows the user's avatar, name, email, role, and active/disabled status.
- Click Disable to ban a user, they will be logged out automatically within 5 seconds, and the user will not be able to log in.
- Click Enable to restore a disabled account.

Step 13: Profile Edit

- Click your avatar/username in the Navbar.
- An overlay opens where you can update your username, email, bio, and profile image.
- For social login users, your Profile Image is automatically synced from your Google or Facebook account.
- Changes are saved to the database immediately.

Step 14: Logout

- Click Logout in the Navbar.
- Session data is cleared from localStorage.
- You are redirected to the Home page.

Step 15: test

- npm run in the terminal to test our project

6. Unique Features

- Real-time discussion updates (no page reload)
- Full discussion thread storage in MongoDB
- Role-based authorization (admin vs user)
- Asynchronous search and filtering
- Docker-based deployment (one command setup)
- Clean UI with structured course browsing
- Social Login including Google and Facebook, profile image is updated with external account profile image